1 Team Round

Problem 1. A six-digit natural number is "*sort-of-decreasing*" if its first three digits are in strictly decreasing order and its last three digits are in strictly decreasing order. For example, 821950 and 631631 are sort-of-decreasing but 853791 and 911411 are not. Compute the number of sort-of-decreasing six-digit natural numbers.

Problem 2. For each positive integer N, let $P(N)$ denote the product of the digits of N. For example, $P(8) = 8$, $P(451) = 20$, and $P(2023) = 0$. Compute the least positive integer n such that $P(n + 23) = P(n) + 23$.

Problem 3. Compute the least integer value of the function

$$f(x) = \frac{x^4 - 6x^3 + 2x^2 - 6x + 2}{x^2 + 1}$$

whose domain is the set of all real numbers.

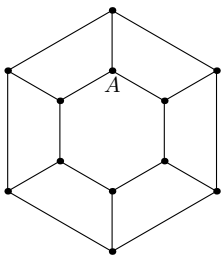
Problem 4. Suppose that noncongruent triangles ABC and XYZ are given such that $AB = XY = 10$, $BC = YZ = 9$, and $\text{m}\angle CAB = \text{m}\angle ZXY = 30^\circ$. Compute $[ABC] + [XYZ]$.

Problem 5. The mean, median, and unique mode of a list of positive integers are three consecutive integers in some order. Compute the least possible sum of the integers in the original list.

Problem 6. David builds a circular table; he then carves one or more positive integers into the table at points equally spaced around its circumference. He considers two tables to be the same if one can be rotated so that it has the same numbers in the same positions as the other. For example, a table with the numbers 8, 4, 5 (in clockwise order) is considered the same as a table with the numbers 4, 5, 8 (in clockwise order), but both tables are different from a table with the numbers 8, 5, 4 (in clockwise order). Given that the numbers he carves sum to 17, compute the number of different tables he can make.

Problem 7. In quadrilateral $ABCD$, $\text{m}\angle B + \text{m}\angle D = 270^\circ$. The circumcircle of $\triangle ABD$ intersects $\overline{CD}$ at point E, distinct from D. Given that $BC = 4$, $CE = 5$, and $DE = 7$, compute the diameter of the circumcircle of $\triangle ABD$.

Problem 8. Suppose that Xena traces a path along the segments in the figure shown, starting and ending at point A. The path passes through each of the eleven vertices besides A exactly once, and only visits A at the beginning and end of the path. Compute the number of possible paths Xena could trace.



Problem 9. Let $i = \sqrt{-1}$. The complex number $z = -142 + 333\sqrt{5}i$ can be expressed as a product of two complex numbers in multiple different ways, two of which are $\left(57 - 8\sqrt{5}i\right)\left(-6 + 5\sqrt{5}i\right)$ and $\left(24 + \sqrt{5}i\right)\left(-3 + 14\sqrt{5}i\right)$. Given that $z = -142 + 333\sqrt{5}i$ can be written as $\left(a + b\sqrt{5}i\right)\left(c + d\sqrt{5}i\right)$, where a, b, c, and d are *positive* integers, compute the lesser of $a + b$ and $c + d$.

Problem 10. Parallelogram $ABCD$ is rotated about A in the plane, resulting in $AB'C'D'$, with D on $\overline{AB'}$. Suppose that $[B'CD] = [ABD'] = [BCC']$. Compute $\tan\angle ABD$.

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